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Model bigger to avoid unrepresentative species distribution models from truncated spatial extents, with the modulating influence of prevalence

Brice B. Hanberry

USDA Forest Service, Rocky Mountain Research Station, Rapid City, SD 57702, United States

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ABSTRACT

As background, one approach for modeling species distributions is to model samples from distributions that are truncated to administrative or ecological boundaries, with predictor variables of climate. My objective was to examine the issue of truncation, which may not generate models that represent species ranges, even inside truncated extents, due to not incorporating the amplitude (i.e., range) of climate values where species occur. For methods, I modeled distributions of 14 wide-ranging tree species in North America and 40 combinations of the tree species truncated within 10 ecological units, using three combinations of extents: both truncated presence and background absence samples (i.e., absences generated from the background extent and hereafter also generally referred to as absences), truncated presence and untruncated absence samples, and untruncated presence and absence samples. Evaluations included model representativeness or similarity to species ranges measured through visualization and the intersection over union score, predicted probabilities in the extents under current and future climate, and climate amplitude values. I also examined how extents combined with equal and unequal modeling prevalence (ratio of presence and absence classes) interacted with ten accuracy metrics, including balanced accuracy (mean of sensitivity and specificity). Presence and absence truncation to ecological units resulted in unrepresentative species distributions (visually, and with a score of 0.1 out of 1), overpredicting presence outside of species ranges and underpredicting presence inside ecological units, due to minimal differentiation in climate variables between presence and absence samples. Mean balanced accuracies for models were 0.83. Presence truncation was extremely accurate, with mean accuracies of 0.99, but models predicted unsuitable climate outside of ecological units within species ranges (score of 0.2). For untruncated samples of 14 tree species, mean accuracies were 0.96 and represented species distributions (score of 0.6). Predicted areas of presence decreased with reduced modeling prevalence, which substantially improved all accuracy metrics for the truncated presence and absence samples but not representativeness of models. In conclusion, representative distribution models, with climate variables, required untruncated presence and absence classes to achieve climate differentiation between presence and absence classes and climate amplitude from the full range of climate values.

1. Introduction

Species distribution models are correlational relationships between species samples and environmental variables that predict probability or suitability of conditions for species presence, resulting in models of species ranges. Climate variables are most commonly applied as predictors for species distribution models and may be the only types of predictors selected for modeling (Mod et al., 2016; Titeux et al. 2016; Fourcade et al. 2018). Due to continuous broad gradients over large extents, climate overall is the most influential type of variable for modeling species distributions and will produce models with high accuracy metrics in terms of classification of samples (Mod et al. 2016; Fourcade et al. 2018; Hanberry 2023a). Species distribution models based on climate delineate the climate conditions of species ranges where species have greater probabilities to occur compared to conditions where species have lesser or no concentrations in occurrences.

However, species distribution models based on climate variables may be representative of species ranges, as defined by models of species distributions that are similar to species ranges, only at scales that cover relatively complete extents of species distributions (Guisan et al. 2025). To produce representative models of suitable climate for species distributions, presence samples need to cover most of the amplitude (i.e.,

E-mail address: brice.hanberry@usda.gov.

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range of values, but here termed as amplitude to differentiate from species ranges) of climate variables within species ranges (Thuiller et al. 2004; Barbet-Massin et al. 2010; Martínez-Freiría et al. 2016; Chevalier et al. 2021; Guisan et al. 2025). Additionally, for species distributions models based on climate, the absence samples from surveys or pseudoabsence samples generated from the background extent need to represent an amplitude of climate conditions where species have lesser probabilities to occur, generally outside of species ranges, incorporating separation from values of climate variables where presence samples are concentrated (Chefaoui and Lobo 2008). Otherwise, absences will contain relatively similar suitable climate conditions as presence samples where the species are more probable to be present, causing underpredictions of presence, even if species cannot completely fill every single location where climate is suitable within or outside ranges (due to factors such as competition, lack of resources, dispersal limitations, disturbances, and extirpations). Truncation of presence samples results in missing information about climate where species occur, reducing predicted extents, and truncation of absence samples results in missing information about climate where species do not occur, which reduces class separability in models.

Nonetheless, one common approach for modeling species distributions is to model species distributions with climate from only portions of ranges, following defined administrative or ecological units that are within portions of species ranges (Beck et al. 2014; Marchi and Ducci 2018; Steen et al. 2021; Andrews et al. 2022; Valavi et al. 2022; Yang et al. 2024; Selvi et al. 2025). Truncated spatial extents and resultant truncated sampling of the climate amplitude in species ranges is defined here as a directly imposed curtailment of full species distributions to administrative or ecological boundaries in spatial extents (Guisan et al. 2025) and the associated measured climate variables, which may deprive species distribution models of completeness or adequacy in climate variable amplitude. Truncating spatial extents in this context is a modeling choice rather than due to less controllable issues such as unknown range extirpations or data limitations.

Truncation of species distributions to spatial extents of interest can reduce representativeness of species distribution models, by instead creating models of partial spatial distributions that may not delineate inherently meaningful units and may not reflect species ranges, even within truncated extents (Chevalier et al. 2021; Bergman et al. 2024; Rojas-Soto et al. 2024). If the intention is to model another biologically meaningful unit than species, such as subspecies, then geographic partitioning of species distributions has a biological basis (Gonzalez et al. 2011; Boyer et al. 2021; Jinga et al. 2021). For truncated ranges, as in extirpated ranges, even if an infinite number of presence samples are supplied as model inputs, those samples will only supply climate information from portions of ranges and not provide the climate amplitude to generate representative species distribution models, that is, models of species ranges (Faurby and Araújo 2018; Hanberry 2023b). This issue is similar to small sample sizes, which indicate that the full climate amplitude was not sampled. Furthermore, species distribution models often are predicted to the future to determine species distributions under climate change, and incomplete coverage and sampling of climate data due to truncation can reduce transferability of models under future climate (Barbet-Massin et al. 2010; Chevalier et al. 2021; Scherrer et al. 2021). Even though this issue has been addressed regularly during past decades, truncated spatial extents continue to be a modeling choice, perhaps in part due to lack of focus on this issue (Thuiller et al. 2004; Barbet-Massin et al. 2010; Hannemann et al., 2016; El-Gabbas and Dormann 2018; Chevalier et al. 2021; Guisan et al. 2025), in favor of issues such as effects of minor errors in databases (Hanberry 2025).

Truncation to spatial extents combined with modeling prevalence, or balance of presence and absence samples for modeling, may influence accuracy metrics, which measure model performance rather than similarity to species ranges. Too little environmental distance between the pseudoabsence and presence samples may result in underpredicted species presence within truncated extents but overpredicted species presence outside of truncated extents (Thuiller et al. 2004; Chefaoui and Lobo 2008). Underpredicted or overpredicted presence areas may be disguised by concurrent unbalancing of samples and then may affect accuracy metrics (Menardi and Torelli 2014). If presence area is overpredicted by the modeling algorithm, due to truncated samples, then increasing the number of absence classes, relative to presence samples, will compensate by overpredicting absence areas (Menardi and Torelli 2014). The modeling algorithm receives samples as two classes of presence and absence and then makes continuous predictions (known as predicted probabilities for the random forests modeling algorithm) for modeled classes based on modeling prevalence to maximize accuracy of the binary classes of presence and absence (Menardi and Torelli 2014). Imbalanced samples then will mask but not correct the underlying issues, arising from truncation, of incomplete information and lack of separation between presence and absence samples. Modeling prevalence alone may have little influence on species distribution models, as long as the threshold for classifying presence and absence and accuracy metrics post-modeling aligns with the modeling prevalence that the modeling algorithm used to predict class probabilities (Hanberry and He 2013). While thresholds may have no biological meaning, modeling prevalence is the meaningful modeling threshold that the modeling algorithm applied for delineating species presence to maximize accuracy metrics, such as balanced accuracy (mean of sensitivity and specificity). Modeling prevalence, and the associated threshold, may alter accuracy metrics without correcting the underlying sampling issue of truncation that limits generation of models that are representative of species ranges.

Truncation to study extents of interest can result in incomplete sampling of the amplitude of climate where species are known to occur and not occur, which in turn may affect species distribution models in terms of representativeness (i.e., similarity to species ranges) and accuracy metrics (such as balanced accuracy, and which measure how well the modeling algorithm modeled and predicted samples), with a modulating influence from modeling prevalence. Here, my objective was to test how truncating spatial extents of species distributions can influence model adequacy, as assessed through multiple evaluation outputs consisting of representativeness measured by visualization and the intersection over union score (i.e., area of overlap relative to total combined area; Jaccard Index; Gilbert 1884) of species distribution models and species ranges, predicted probabilities within and outside of modeled ecological units (i.e., ecoregions; Olson et al. 2001; Fig. 1) under current and future climate, climate variable amplitude, and accuracy metrics, and also the modulating influence of modeling prevalence on accuracy metrics, which evaluate model agreement with the input samples rather than representativeness of species distribution

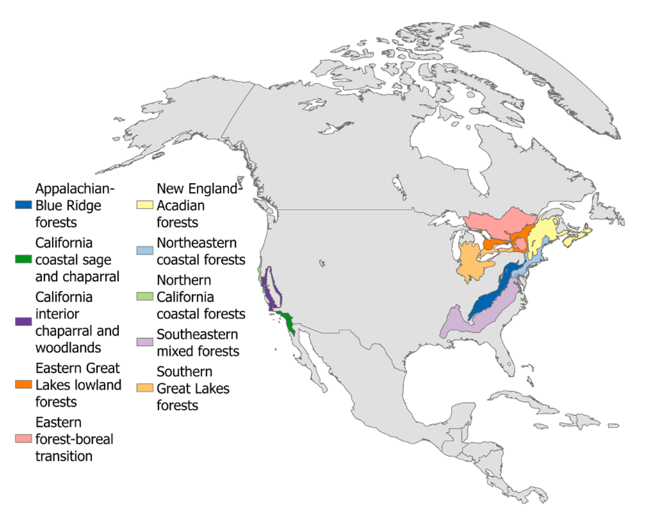

Fig. 1. Ecological units for truncation of species distributions.

models. For truncation, I modeled the distributions of 14 wide-ranging North American tree species and 40 combinations of the 14 tree species truncated within 10 ecological units, with both truncated presence and absence samples, truncated presence and untruncated absence samples, and untruncated presence and absence samples. As an addition, presented briefly in the supplementary material is the option to generate target-group absence samples specifically in locations with higher intensities of occurrences for all species of similar taxa to cancel out survey accessibility and observer effort (Phillips et al. 2009; Barber et al. 2022; Supplementary file 1).

2. Methods

2.1. Overview

My objective was the examine effects of truncation, with balanced and imbalanced modeling prevalence, on species distribution models (Supplementary file 2; ODMAP, Zurrell et al. 2020). Models were from either truncated georeferenced records and absence samples in ecological units, from truncated georeferenced records in ecological units and absence samples from the North American extent, or from untruncated georeferenced record and absence samples from the North American extent. Other modeling selections, besides sampling extents and modeling prevalence, were held constant. The modeled tree species were well-sampled, with wide-ranging distributions in North America excepting one species. To achieve balanced modeling prevalence, or ratio of presence and absence samples, I generated an equal number of random samples from the background study extent as the number of presence samples, which was the basis for predicting presence and metrics at a 0.5 predicted probability threshold (Hanberry and He 2013), which is the default threshold (caret R package; Kuhn 2008; R Core Team 2024). In addition, I imbalanced the modeling prevalence to 0.33, as an exception to the modeling framework, but without adjusting the standard threshold of 0.5 for prediction of presence, including in accuracy metrics to demonstrate the effect of unadjusted, imbalanced prevalence. Predictor variables were 13 climate variables, with values extracted at the species presence and absence coordinates, for all models.

I applied the random forests classifier, an ensemble nonlinear model, with the one uninfluential hyperparameter selected through 10-fold cross-validation, repeated three times (automated in caret R package; Kuhn 2008; Probst et al. 2019; R Core Team 2024). General model insensitivity occurs whether one variable or all variables for the hyperparameter of number of randomly selected predictors considered at each split in a tree, at least for important variables such a climate (Probst et al. 2019). I modeled one time with 75% of the samples to determine how well the models predicted the withheld 25% of samples, to test for both overfitting and accuracy metrics (see accuracy metrics section below; createDataPartition and confusionMatrix functions in caret R package; Kuhn 2008; R Core Team 2024). I modeled a second time with all samples to map predictions. Assessment was by visualization, the intersection over union score relative to range maps (i.e., area of overlap relative to total combined area; values bounded from 0 to 1; values > 0.5 are generally good, depending on application; Jaccard Index; Gilbert 1884), mean predicted probabilities in the North America extent and within the ecological units under current and future climate, range of climate values of the presence and absence samples, and accuracy metrics.

2.2. Study extents

The untruncated modeling extent was the North American continent, encompassing the subcontinent of Central America and the Caribbean, excluding smaller islands. The spatial extents for the truncated species distributions were 10 ecological units in North America (ecoregions; Olson et al. 2001; Fig. 1). These ecological units were used to truncate species georeferenced records and background absence samples. There were 40 ecological unit and tree species combinations, consisting of two to five of the 10 ecological units for each of the 14 tree species.

2.3. Species data

The Global Biodiversity Information Facility (GBIF; Global Biodiversity Information Facility 2024) is a centralized data repository of georeferenced records of species observations. Basic data processing removed records without coordinates and duplicates by species-coordinate-year and selected locational uncertainty based on distances (keeping distances <1 km), collection years 1990 to 2020, basis of record (i.e., preserved specimens, human observations, or observations), and terrestrial coordinates located in the North American continent (GBIF Occurrence Download https://doi.org/10.15468/dl.k59cjh accessed on 2 August 2024). I identified tree species that had at least 1000 unique locations and coordinate uncertainty <1 km during 1990 to 2020 in two or more ecological units in the North American continent (Fig. 1). This resulted in 14 tree species: *Acer negundo* L., *Acer rubrum* L., *Acer saccharum* Marshall, *Fagus grandifolia* Ehrh., *Juniperus virginiana* L., *Liquidambar styraciflua* L., *Liriodendron tulipifera* L., *Pinus strobus* L., *Platanus occidentalis* L., *Quercus agrifolia* Née, *Quercus alba* L., *Quercus rubra* L., *Sequoia sempervirens* (D.Don) Endl., and *Tsuga canadensis* (L.) Carrière. Both unthinned and thinned (i.e., removal of local clustering) samples, or inclusion and exclusion of clustered urban samples, will produce representative species distributions models, albeit slightly different approximations (Hanberry 2025). I did not thin clustered samples, because thinning to small sample sizes also can result in overpredicted areas (i.e., unrepresentative models, Hanberry 2025) and thinning typically does not improve models (Lamboley and Fourcade 2024).

2.4. Climate variables

Following the common approach of selecting climate data that may occur after tree observations, to prevent divergence from common practice, I extracted debiased and downscaled recent 30-year climate during 1981–2010 (resolution 30 ′; CHELSA climate; Karger et al. 2017, 2018). I selected and clipped to North America 13 bioclimatic variables that quantify temperature and precipitation at annual, seasonal, and monthly intervals,: mean annual temperature, maximum temperature of the hottest month, minimum temperature of the coldest month, mean temperature of the coldest and hottest three months, mean temperature of the driest and wettest three months, annual precipitation, precipitation during the driest month and driest three months, precipitation during the wettest month, and precipitation during the coldest and warmest three months. To demonstrate effects of truncated species distribution modeling on future predictions, I modeled predictions to future climates during 2071–2100. Future climate projections included a range of climate for North America from the moderate SSP3–7.0 scenario: GFDL-ESM4 (GFDL; Geophysical Fluid Dynamics Laboratory, USA) that projected an increase of 4.3 °C mean annual temperature, IPSL-CM6A-LR (IPSL; Institut Pierre Simon Laplace, France) with an increase of 6.3 °C mean annual temperature, and UKESM1–0-LL (UKESM1; Met Office Hadley Centre, UK) with an increase of 8.8 °C mean annual temperature in North America, excluding Greenland.

Regarding number of climate variables, I did not reduce the 13 climate variables, to avoid exclusion of relevant, proximal variables for modeling species ranges with climate (Elith and Leathwick 2009; Hanberry 2024) and to hold comparisons as equal as possible. Because sampling extents were being varied, the number of climate variables (13) and other modeling components were held equal (ceteris paribus). Correlated climate variables are not an issue for predictions, in linear or nonlinear models, only for the standard error of estimates in linear models (Hanberry 2024). Correlation assessments have demonstrated that inclusion of important, relevant climate variables to models will avoid potential bias (Hanberry 2024).

2.5. Species distribution modeling

I extracted climate values at the species georeferenced records and background absence coordinates, generated randomly within the background extents of either ecological units or North America (terra package, Hijmans et al. 2026; R Core Team 2024). To examine truncation with balanced modeling prevalence, I modeled 40 combinations of 14 tree species with truncated presence and absence samples within 10 ecological units (two to five ecological units per species), 40 combinations of 14 tree species with truncated presence within 10 ecological units and absence samples from the North American background extent, and 14 tree species models with presence samples from throughout species ranges and absence samples from the North American background extent. To achieve balanced modeling prevalence, or ratio of presence and absence samples, I generated an equal number of random background absence samples from the background study extent (i.e., either the ecological units or North America) as the number of presence samples. The modeling prevalence was 0.5, a modeling threshold that was the basis for predicting presence by the modeling algorithm and metrics at 0.5 predicted probability (Hanberry and He 2013), which is the default threshold (caret package, Kuhn 2008; R Core Team 2024). For the 14 tree species with untruncated samples and only one ecological unit per tree species for truncated samples, I modeled with imbalanced samples, at twice the number of absence classes as presence classes (modeling prevalence to 0.33). I maintained the threshold at 0.5 to demonstrate the effect of unadjusted, imbalanced prevalence.

For modeling of each species, I applied the random forests classifier, an ensemble nonlinear model, to model presence and background bsence samples based on climate variables and then predict presence and absence classes at the 0.5 predicted probability threshold (caret package; Kuhn 2008; R Core Team 2024). Model configuration (hyperparameter) tuning was automatically selected through random 10-fold cross-validation, repeated three times, but default hyperparameters for the random forests algorithm are robust or have little influence on models (caret package; Kuhn 2008; Probst et al. 2019). In the caret package, the tuning parameter for random forests is the number of randomly selected predictors considered at each split in a tree, which typically will be two variables with relevant climate variables, or close to the square root of the total number of variables, a random forests default; number of trees (500) and node size (1) are default values (caret package; Kuhn 2008; Probst et al. 2019). Withheld samples, known as out-of-bag samples, for each tree providing internal unbiased accuracy without a need for a separate testing dataset (Breiman 2001a, b). I modeled one time with 75% of the samples to determine how well the models predicted the withheld 25% of samples, for accuracy metrics (see accuracy metrics section below; createDataPartition and confusionMatrix functions in caret package; Kuhn 2008). To test for overfitting through discrepancies between training and testing performance, I extracted accuracy of out-of-bag samples, training samples, and testing samples. I modeled a second time with all samples to map predictions.

2.6. Measurable outcomes

Visualization of predicted areas of species presence compared to species ranges, or alternatively at least untruncated sampled areas, shows whether species distribution models are representative. As a simplified, objective metric, I calculated the area of intersection over union score (values bounded from 0 to 1; values $>$ 0.5 are generally good; Jaccard Index; Gilbert 1884), for comparison between predicted areas by species distribution models and species range maps (https://doi.org/10.5281/zenodo.15468153; Little 1971). This direct comparison to range maps will be the same for all sampling approaches, without effects of species distribution models and differential effects of truncation. I also quantified mean predicted probabilities for the three combinations of modeling extents in the North America extent and within the ecological units and under current and future climate. In addition, I displayed the range of climate values from presence and absence samples for the three modeling extent combinations.

For accuracy metrics, accuracy is a necessary but not sufficient condition for representative species distribution models. Accuracy indicates whether the modeling algorithm can differentiate presence and absence samples based on predictor variables but not whether the input samples represent species ranges. For all models, I selected three straightforward accuracy metrics of balanced accuracy (mean of sensitivity and specificity), sensitivity or true positive rate, and specificity or true negative rate for 25% withheld samples (using confusionMatrix function in caret R package, with a modeling prevalence of 0.5 and a threshold of 0.5 between absence and presence classes; Kuhn 2008; R Core Team 2024). To test for overfitting, I applied the most common and objective metric of comparison between accuracy of training data and withheld samples, because the problem with overfit models is lack of transferability, resulting in poor accuracy for samples not used in modeling (Warren and Seifert 2011). Additionally, I recorded the withheld accuracy of out-of-bag error that occurs internally in random forests.

For the subset of imbalanced models and the equivalent models with balanced samples (i.e., only one model for each of the 14 tree species for the three extents), I examined a range of metrics: balanced accuracy, sensitivity, specificity, true skill statistic (same as Youden's index J), AUC (area under curve), kappa, phi coefficient (Matthew’s correlation coefficient), Brier score (similar to mean squared error in that small values have less divergence in predictions), negative predicted value, and positive predicted value (Kuhn 2008; Liu et al. 2011; Chicco et al. 2021; Probst, 2021; Kuhn et al., 2024). Metrics were calculated in two of three packages for confirmation purposes (confusionMatrix function in caret R package, Kuhn 2008; conf_mat, roc_auc, brier_class functions in yardstick R package, Kuhn et al., 2024; AUC, brier, KAPPA, MCC functions in measures R package, Probst 2021). For imbalanced modeling prevalence, I maintained the threshold at 0.5 between absence and presence classes to demonstrate the effects of unadjusted, imbalanced prevalence. To compare metrics for the six combinations of three combinations of modeling extents and two modeling prevalences, I also correlated values (including for the inverse of the Brier score; ggcorrplot R package; Kassambara, 2023).

3. Results

3.1. Visualization under current and future climate

Visualization of model predictions illustrated the issues of truncated distributions, which were conveyed to future predictions (Figs. 2–3). For truncated presence and absence samples for 40 ecological unit and species combinations, predictions generally overpredicted presence where the species were absent and underpredicted presence where the species were present, with variable outcomes depending on the ecological unit and species combinations. Conversely, for truncated presence samples and untruncated absence samples from North America for ecological unit and species combinations, predictions were limited precisely to the ecological units and immediate surroundings. For any truncated samples, predictions were not representative of complete species distributions from these wide-ranging species, except for *Sequoia sempervirens,* which has a restricted distribution. The species distribution models from untruncated samples of 14 tree species displayed the full extent of these wide-ranging, well-sampled species (i.e., relative to observed ranges from untruncated samples).

3.2. Intersection over union score

For modeled truncated presence and absence samples in ecological unit and species combinations, the mean intersection over union score

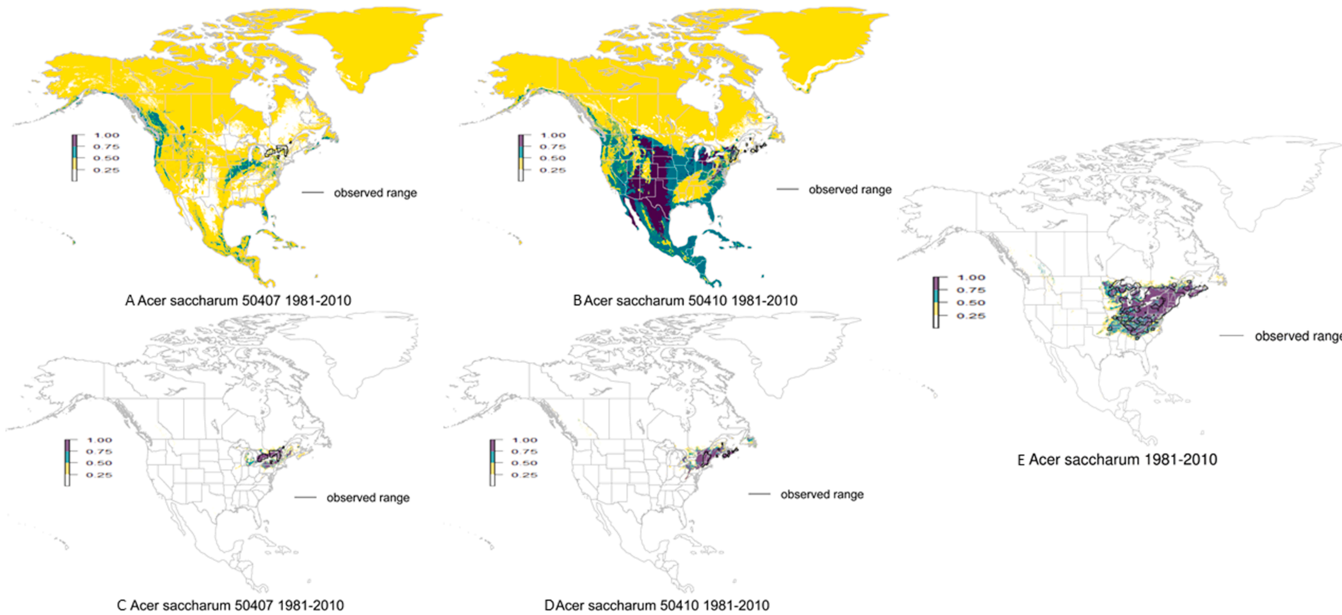

Fig. 2. Example predictions of probabilities for the two ecological units (50,407 = Eastern Great Lakes lowland forests, 50,410 = New England-Acadian forests) where *Acer saccharum* was abundant, from modeling truncated presence and absence samples (A, B), truncated presence and untruncated absence samples (C, D), compared to untruncated presence and absence samples, which represents the range of *Acer saccharum* (E). Observed range (black outline) indicates location of species presence samples.

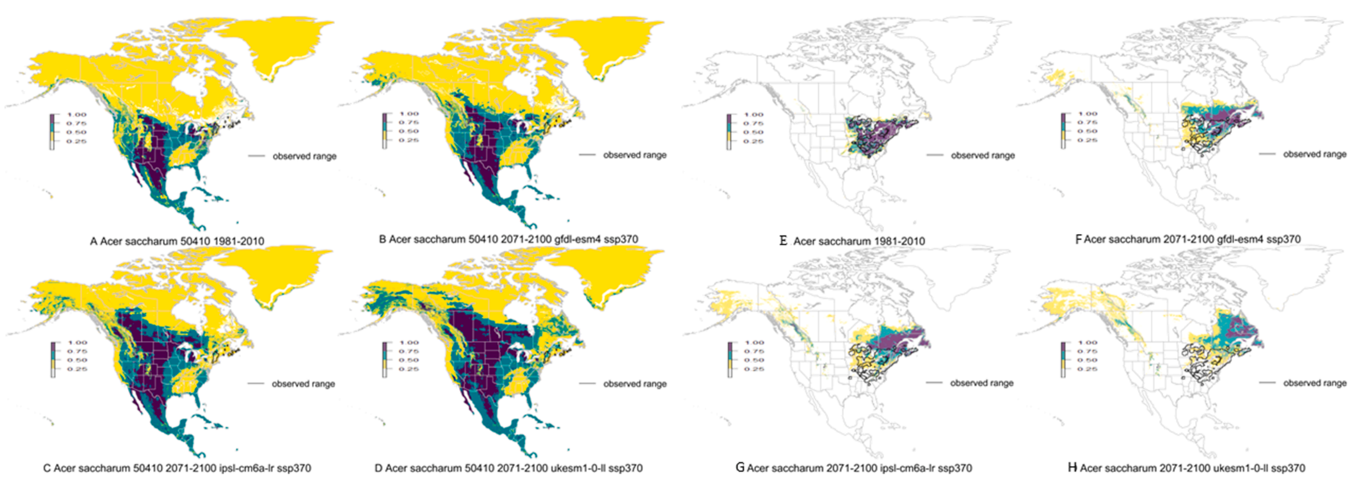

Fig. 3. Example predictions from modeling truncated presence and absence samples under current climate (1981–2010; A) and three future climate projections (2071–2100; B–D) for one ecological unit (50,410 = New England-Acadian forests) where *Acer saccharum* was abundant. Observed range (black outline) indicates location of species presence samples. Example predictions from untruncated presence and absence samples under current climate (1981–2010; E) and three future climate projections (2071–2100; F–H) for *Acer saccharum*. Observed range (black outline) indicates location of species presence samples.

with ranges was 0.1 (out of a maximum score of 1). For truncated presence samples and untruncated absence samples, the mean intersection over union score was 0.2. For untruncated presence and absence samples of tree species, the mean intersection over union score was 0.6, generally acceptable, due to particular overprediction for *Sequoia sempervirens* (intersection over union score was 0.07).

3.3. Mean predicted probabilities under current and future climate

Mean predicted probabilities (Fig. 4) quantified prediction mapping under current climate (1981–2010) and three future climate projections (2071–2100). For models from truncated presence and absence samples in ecological unit and species combinations, mean predicted probabilities were high where the species were recorded as absent for the North American extent and mean predicted probabilities were low where the species were known to be present in the modeled ecological units. For three future climate projections, mean predicted probabilities increased under future climate in ecological units relative to mean predicted probabilities where species occur now under current climate. Conversely, models from truncated presence and untruncated absence samples had low predicted probabilities outside of the modeled ecological units under current climate, including within ranges, and high predicted probabilities within ecological units that decreased to low predicted probabilities under future climate. Models from untruncated samples had low predicted probabilities outside of all modeled ecological units under current climate and high predicted probabilities within all ecological units that decreased to moderate predicted probabilities under future climate, as species distributions shifted north.

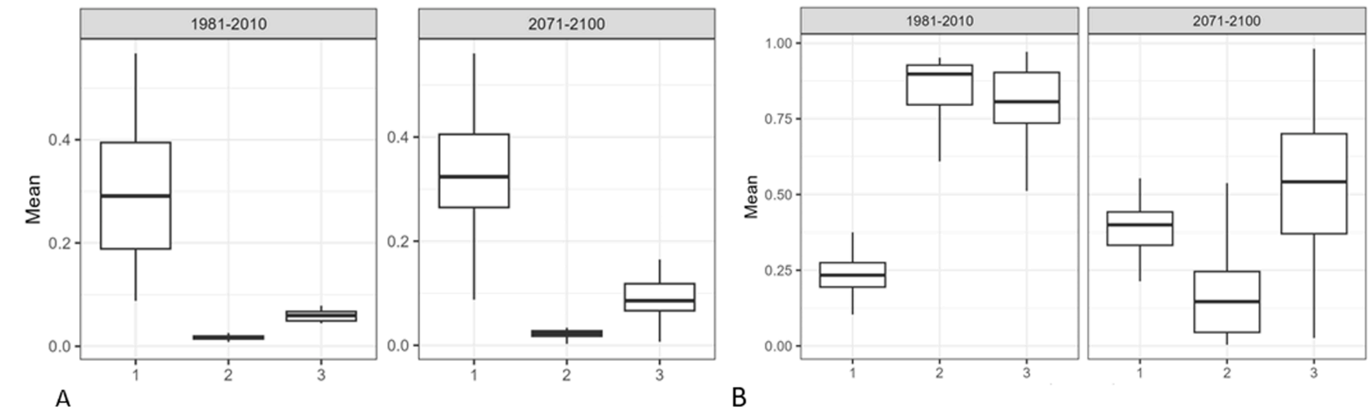

Fig. 4. Mean predicted probabilities of climate suitability by models of truncated presence and absence samples (1), truncated presence and untruncated absence samples (2), and untruncated presence and absence samples (3) under current climate (1981–2010) and three future climate projections (2071–2100) in North America (A) and in modeled ecological units (B).

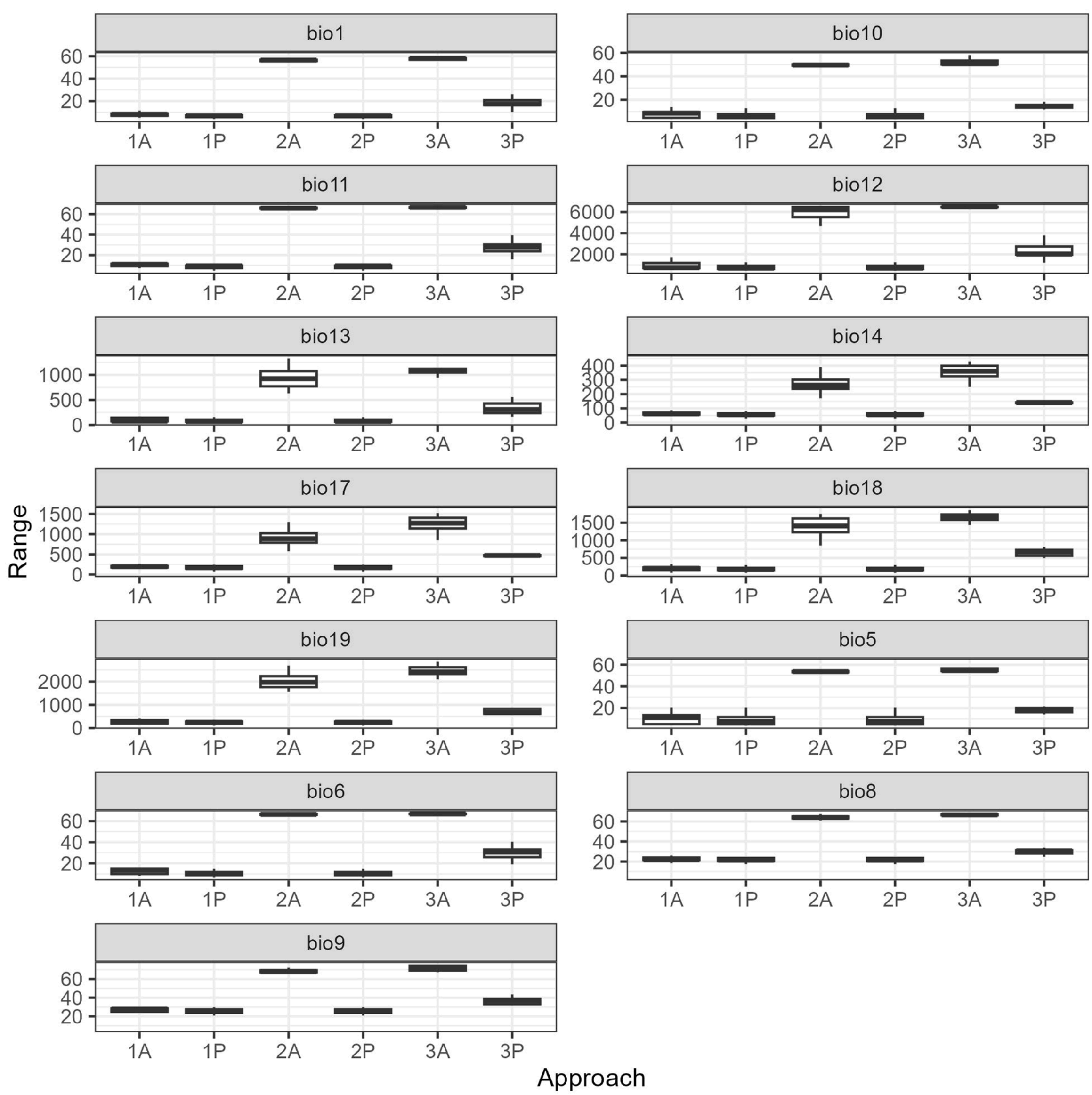

Fig. 5. Amplitude (range) of values for 13 climate variables for presence and absence samples of the three combinations of modeling extents (1A = truncated absence samples, 1P = truncated presence samples, 2A = truncated absence samples, 2P = untruncated presence samples, 3A = untruncated absence samples, 3P = untruncated presence samples).

3.4. Climate predictor range of values for truncated and untruncated samples

The range of values for the 13 climate variables demonstrated the issues of truncated presence and absence samples (Fig. 5). Values of climate variables did not have any separation between the presence and absence samples for models from truncated presence and absence samples. The amplitude of values for climate variables was limited for truncated presence samples, unlike the untruncated presence samples.

3.5. Accuracy metrics for truncated and untruncated samples

For modeled truncated presence and absence samples in ecological unit and species combinations, mean balanced accuracies of withheld samples for models were 0.83 (sensitivity = 0.83, specificity = 0.83), which was a decrease of 0.0023 and 0.0044 from accuracy of training data and out-of-bag error, respectively. For truncated presence samples and untruncated absence samples, mean accuracies of withheld samples for models were 0.99 (sensitivity = 0.996, specificity = 0.99), which was a decrease of 0.0008 from accuracy of both training data and out-of-bag error. For untruncated presence and absence samples of 14 tree species, mean accuracies of withheld samples for models were 0.96 (sensitivity = 0.98, specificity = 0.94), which was a decrease of 0.0013 and 0.0018 from accuracy of training data and out-of-bag error, respectively.

3.6. Accuracy metrics for imbalanced modeling prevalence

For truncated presence and absence samples, the accuracy metrics increased for decreased modeling prevalence (of twice the number of absence classes as presence classes), relative to balanced prevalence, except for the Brier score, a difference metric, which correspondingly decreased (threshold of 0.5; Fig. 6). Increasing the number of absences forced the model to reduce overpredictions of the presence class. For truncated presence and untruncated absence samples, because the predictions were so precise ($\geq$ 0.98), all of the accuracy metrics essentially remained stable. For untruncated samples, the accuracy metrics shifted slightly to reflect emphasis on predictions of the absences, resulting in increased specificity (+0.020) and decreased sensitivity (−0.007).

Accuracy metrics were similar and also responded similarly to the three combinations of extents and two modeling prevalences (Fig. 7). Sensitivity and specificity were most dissimilar in response (r = 0.78). Balanced accuracy, Kappa, Phi, and true skill statistic were most similar in response, and Kappa, Phi, and true skill statistic were identical, or nearly so, in values. The AUC and Brier metrics had little range in values, with compressed very high or very low values, respectively.

4. Discussion

4.1. Unrepresentative models from truncated samples

A question to consider for species distribution modeling is how representative species distribution models are for truncated samples, as a modeling decision (Thuiller et al. 2004; Barbet-Massin et al. 2010; Martínez-Freiría et al. 2016; Chevalier et al. 2021; Guisan et al. 2025). The clear findings from the study are that approaches to constrain samples, whether presence or absence samples, will produce

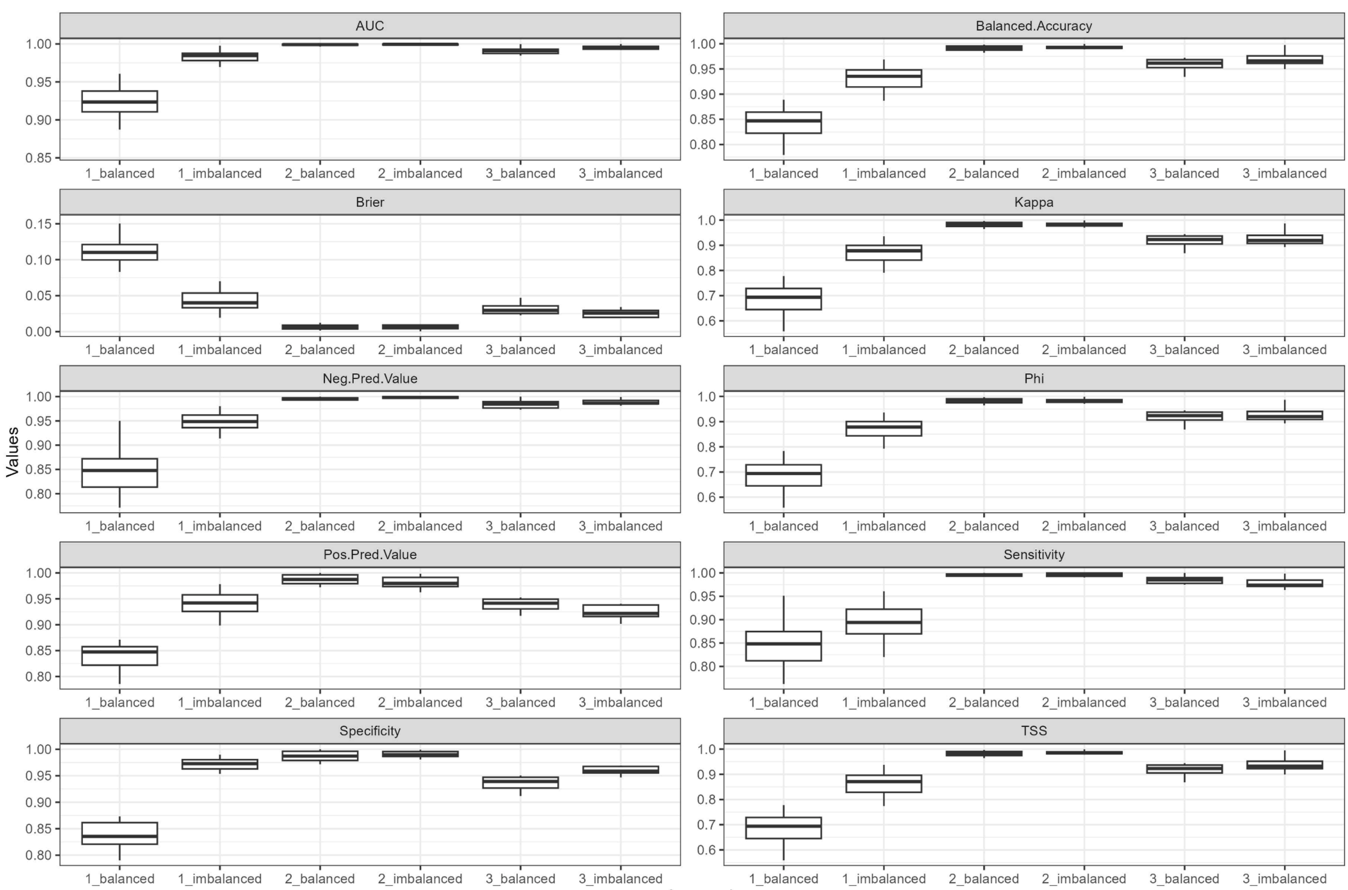

Fig. 6. Accuracy metrics of AUC, accuracy, Brier score, Kappa, negative predicted value, phi coefficient, positive predicted value, sensitivity, specificity and true skill statistic for one ecological unit per tree species for models of truncated presence and absence samples (1) and truncated presence and untruncated absence samples (2) and 14 tree species for untruncated presence and absence samples (3) and balanced samples or imbalanced samples (twice the number of absence classes as presence classes).

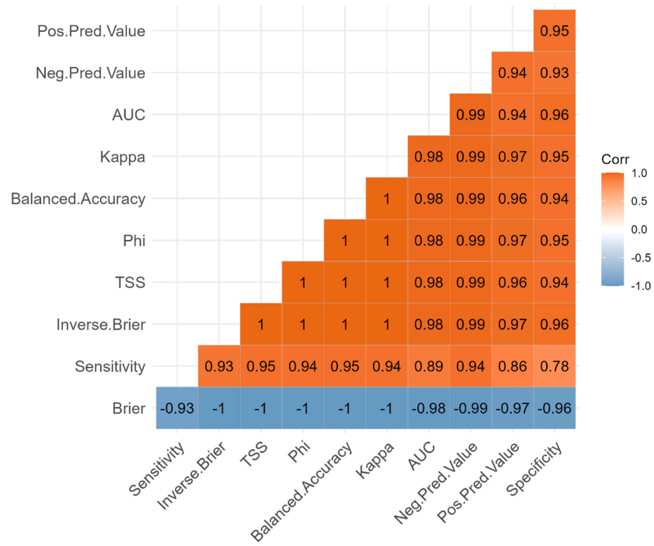

Fig. 7. Correlation matrix of accuracy metrics (with inverse Brier score).

unrepresentative species distributions for models based on climate variables, as measured by visualization, intersection over union scores relative to ranges, mean predicted probabilities, and climate amplitude values (Thuiller et al. 2004; Guisan et al. 2025). Truncation of both presence and absence samples provided the least amount of information for the modeling algorithm to differentiate classes based on climate, resulting in models that predicted widespread presence outside of ecological units and species ranges where species are known to occur. In contrast, truncated presence samples and untruncated absence samples enabled precise separation between suitable and unsuitable predictions of species presence, generating extremely specific models of modeled ecological units, with underpredictions of climate suitability outside of ecological units within species ranges. Under future climate within ecological units, the truncated approaches likewise either predicted greater probabilities than under current climate because models shifted into ecological units from outside of ecological units or low probabilities because truncated portions of species distributions shifted out of ecological units, without replacement by more southern untruncated portions of species ranges. As a solution to these issues, untruncated presence and absence samples provided the climate amplitude and climate value differentiation necessary to model species distributions, with high predicted probabilities where species occurred, low probabilities where species were unknown, and moderate probabilities under future climate within ecological units. In terms of transferability, models were not overfitted to training data, according to the objective measurement of discrepancies between training and testing performance (Warren and Seifert 2011), and continental models based on climate variables, which are correlated in time, are similar under past and future climate (Hanberry 2023c, 2034). While others have researched this issue, results sometimes have appeared ambiguous due to varying study extents, eluding recommendations, which indicates how serious and pervasive truncation is for species distribution modeling (Hannemann et al., 2016; El-Gabbas and Dormann 2018; Bergman et al. 2024).

Truncated samples were not representative of species distributions due to lack of information about amplitude (range) of climate variables of the species distributions. If climate for species occurrences is truncated to study extents relatively smaller than species ranges, modeling algorithms do not have the input information that the species distributions, and associated amplitude of climate variables of the species distributions, extend beyond the provided samples. Therefore, for truncated presence samples, the modeling algorithm predicted low probabilities outside of ecological units to accurately model the truncated climate data. Despite high accuracy, due to models correctly limited to the smaller areas of the ecological units rather than entire ranges, models were not representative of the species distributions because the models did not transfer outside of the ecological units, where the species ranges occurred.

Moreover, if modeling algorithms receive absence samples from truncated climate data, overlapping with presence samples, algorithms may not be able to differentiate between suitable and unsuitable climate. Therefore, for truncated presence and absence samples, the modeling algorithm predicted high probabilities outside of the ecological units and low probabilities inside ecological units, as also was depicted by Thuiller et al. (2004). Lack of differentiation between presence and absences samples is likely to be amplified by intentionally targeting absence samples to overlap with presence samples (Phillips et al. 2009; Barber et al. 2022; Supplementary file 1). Indicating a technical issue in modeling to maximize predictions of presence areas where species occur, these models were not representative within or outside of the ecological units, and climate suitability for species increased over time in ecological units. Although true future distributions are unknown, the expectation ecologically and in modeling is that species distributions will shift away from current locations to maintain climate conditions under future climate, as documented under past climate change (Williams et al. 2002, 2013).

A full amplitude of absence samples from continental scales, or at least large extents of continents that exhibit a range of climate, will provide the modeling algorithm with values of climate where species are less probable to occur, avoiding issues arising from missing or overlapped climate amplitudes of presence and absence samples. Indeed, model predictions may be improved by completely excluding absence samples from species ranges (Hanberry 2023b), albeit not necessary or perhaps even warranted (Stokland et al. 2011). Others have recommended a more moderate background extent for background absence sample generation to prevent inflation of accuracy metrics (VanDerWal et al. 2009). Given that the intention of modeling and measurement of model accuracy is to produce accurate models, greater accuracy is a feature, not an unintended flaw, to demonstrate that more information was conveyed to the modeling algorithm about climate amplitudes where species do not occur, as opposed to withholding information from the modeling algorithm to depress accuracy and impede differentiation between classes. Conversely, artificially creating separation between values of climate for presence and absence samples, for example, by imposing geographic buffers, likely will cause the modeling algorithm to have trouble differentiating classes for intermediate values. Absence samples at the continental scale will allow the modeling algorithm access to a full range of climate amplitudes, particularly critical for modeling of distributions under future climate. Therefore, a clear recommendation is to model entire species distributions, not limited to ecological or administrative units, with absence samples from continental scales, or near continental scales, to avoid unrepresentative species distributions.

4.2. Accuracy metrics

Modeling algorithms can increase accuracy by allocating predictions proportional to the modeling prevalence ratio, that is, predicting more absence classes when samples are imbalanced to the absence class (Menardi and Torelli 2014). Accuracy of models from truncation of both presence and absence samples was improved by decreased modeling prevalence. All the accuracy metrics were affected by changing prevalence similarly, including the true skill statistic, a rescaled version of balanced accuracy (Somodi et al. 2017; Leroy et al. 2018). Due to models that overpredicted presence overall, but underpredicted within ecological units, from truncated presence and absence samples, all accuracy metrics improved considerably with prevalence that was imbalanced to absence samples, which caused the modeling algorithm to increase predictions of absence. However, the other two combinations of modeling extents had high accuracy metrics and contained separation between presence and absence samples, resulting in minimal effects on accuracy by prevalence.

Guidelines about accuracy metrics are arbitrary without the modeling context and additional evaluations. Nevertheless, for species distribution models based on climate variables, a binary classification that is extremely accurate, a variety of accuracy metrics are correlated and appropriate but probably should be at minimum 0.90 to be accurate, depending on scaling of the accuracy metric (e.g., greater minimum values for AUC, resulting in values compressed to a small range). Any values below about 0.85 indicate issues that should be examined more closely for species distribution models based on climate.

For species distribution modeling, evaluations of performance other than accuracy metrics are necessary, such as assessments of prediction maps compared to range maps, predicted probabilities, and climate amplitude of the samples. The most accurate models were produced by truncating presence samples to ecological units smaller than species ranges. While these models accurately reflected the ecological units, the models were inaccurate compared to known species ranges. Truly accurate models are representative of species distributions, and not simply models with high accuracy metrics.

4.3. Using truncated models

Characteristics of species ranges may lessen effects of truncated samples, which can be assessed for climate amplitude (range of values). Spatial units may contain a wide climate amplitude due to elevational gradients, with mountains and plains, or span wide latitudes. Modeling of species that have clear separation in space due to climate also will improve models. However, species that occur along elevational gradients, with the implication of climate influence, may be responding to disturbances and land uses (Brandegee 1899; Town 1899; Leiberg 1904; Gruell 1980). Rare species with extremely restricted ranges may have narrow climate amplitudes that fit within modeling units, as indicated by more representative models from truncated samples of *Sequoia sempervirens,* which had a limited range relative to other modeled tree species. But, species with very limited ranges may not be appropriate candidate species for climate-driven species distribution models, given that small distributions probably are imposed by other factors than climate and the species likely can tolerate a greater range of climate than indicated by distributions.

If the truncated species distribution models are not intended to predict in space (other extents) or time (other climates), then the results may be valid as 'here and now' models, particularly if climate variables are not predictors in models or if climate variables are applied for the full species distributions in spatially-nested species distribution models (Guisan et al. 2025). Climate variables typically will override the influences of other variables. Other variables may be influential at local scales and critically, nearly the full range of potential values for non-climate variables may be present in small extents. Summary statistics for variables where the species presence samples occur and where absence samples occur will help answer whether separation in predictor values allows for representative models.

Across studies, besides the challenges of comparing models when many factors are different than the specific ones being considered, one or two choices can produce conflicting results, resulting in lack of agreement and advancement of best modeling practices. Models that are uncertain and unrepresentative in predictions, owing to lack of climate amplitude in species samples and perhaps additional incomplete separation in climate between presence and absence samples, probably will not be appropriate products overall, at least for wide-ranging species. Truncated samples will produce unrepresentative and unreliable models because the full climate amplitude is not included in models, regardless of high accuracy metrics. Furthermore, models that do not represent species distributions, and even with acknowledgement of poor or potentially poor models (Beck et al. 2014; Valavi et al. 2022), seemingly do not provide secure foundations for inference about species distribution models or more generally, different performance aspects of species distribution models (Moudrý et al. 2024).

5. Conclusions

Species distribution models from truncated samples, modeled from climate predictors with limited amplitude in values, overall do not represent species ranges. Truncated presence and absence samples resulted in minimal separation of climate variable values between presence and absence classes and truncated climate amplitude (i.e., range of values) for the species distributions. Truncated presence and absence samples generated unconstrained predictions of presence outside of ranges and underpredicted presence within the modeled ecological units. Truncated presence samples lacked the full climate amplitude for the species, generating predictions constrained to the ecological units, which were not correct representations of species ranges. Despite models of truncated predicted presence areas that did not represent species ranges, the truncated models may be extremely accurate, in terms of accuracy metrics, based on the truncated information provided to the modeling algorithms. Truncated samples consistently produced unrepresentative distributions, with the exception of one tree species with a restricted distribution, and may be the greatest correctable issue for modeling species distributions with climate predictors. Presence samples need to be representative of species ranges and absence samples need to be representative of a wide range of conditions outside of species ranges. A clear recommendation is to model bigger, at continental scales, for representative species distributions based on climate.

CRediT authorship contribution statement

Brice B. Hanberry: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.envc.2026.101551.

Data availability

The untruncated models with balanced samples are available at 10.5281/zenodo.17180183. The truncated models are available upon request.

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